

Francisco Richter Mendoza

Researcher · Mathematical Engineer

✉ richtf@usi.ch · 🌐 francisorichter.github.io · 🐙 francisorichter · 🌐 francisorichter

PROFILE

I work on inference for complex systems: what a partial, non-random observation of a system determines about the process that generated it, and what it leaves undetermined. My current work covers identifiability and partial identification in macroevolutionary models, causal invariance under distribution shift without environment labels, and graphical models of interaction networks. I collaborate directly with biologists, astronomers and statisticians, turning open-ended questions in their fields into computational ones.

EDUCATION

- 2015 – 2021 **PhD** University of Groningen, The Netherlands
Thesis: *Inferring the drivers of species diversification: using statistical network science*. Supervised jointly by E. C. Wit (Probability & Statistics) and R. S. Etienne (Theoretical & Evolutionary Community Ecology).
- 2013 **Mathematical Engineer** Univ. Técnica Federico Santa María, Chile
Thesis: *Mathematical modelling applied to photometric correction in astronomical observations*.

EXPERIENCE

- Feb 2026 – **Scientific Staff (part-time)** USI Università della Svizzera italiana, Lugano
Statistical network science, biostatistics and complex systems; modelling of species diversification and phylogenetic inference.
- 2023 – 2026 **Lecturer** USI Università della Svizzera italiana, Lugano
Probability & Statistics, Numerical Computing, Stochastic Methods, Ordinary Differential Equations, Data Science and Social Network Analysis (Bachelor INF; Master IA/CS).
- 2021 – 2023 **Postdoctoral Researcher** USI Università della Svizzera italiana, Lugano
A general phylogenetic inference framework for species diversification; graphical models of microbiome interactions across related environments.
- 2020 **Visiting Researcher** USI Università della Svizzera italiana, Lugano
COST STSM grant (Action CA15109): statistical network science, in a Groningen–USI collaboration.
- 2012 – 2015 **Senior Research Associate & Data Scientist** Evalueserve, Viña del Mar, Chile
Data science and operational risk modelling for financial clients.
- 2009 – 2015 **Lecturer & Research Assistant** Univ. Técnica Federico Santa María, Chile
Lectured *Matemáticas I* (2015); artificial-intelligence methods for ALMA observational scheduling (2009–2012).

2012 – 2013

Research Assistant ESO European Southern Observatory, Santiago, Chile
Mathematical methods for photometric correction on the Very Large Telescope (VLT).

PUBLICATIONS

- 2026 Where the tree of life is empirically resolved, and where it is not: an open atlas of species-level phylogenies and their archival uncertainty
F. Richter
bioRxiv, 2026. doi : 10.64898/2026.05.31.729127
- 2025 Neural causal regularization: extending causal invariance to deep models
F. Richter, K. Rigana, E. Wit
Statistics for Innovation I, 97–102. Springer Nature, 2025
- 2024 Random graphical model of microbiome interactions in related environments
V. Vinciotti, E. C. Wit, F. Richter
Journal of Agricultural, Biological and Environmental Statistics, 2024
- 2021 Detecting phylodiversity-dependent diversification with a general phylogenetic inference framework
F. Richter, T. Janzen, H. Hildenbrandt, E. C. Wit, R. S. Etienne
bioRxiv, 2021
- 2020 Introducing a general class of species diversification models for phylogenetic trees
F. Richter, B. Haegeman, R. S. Etienne, E. C. Wit
Statistica Neerlandica 74, 261–274. Wiley, 2020
- 2021 Microbiome and environment explain the absence of correlations between consumers and their diet in Bornean microsnails
K. P. Hendriks, ..., F. J. Richter Mendoza, M. Schilthuizen, R. S. Etienne
Ecology 102.2, e03237, 2021
- 2021 Plant diets of land snail community members are similar in composition but differ in richness
K. P. Hendriks, ..., F. J. Richter Mendoza, M. Schilthuizen, R. S. Etienne
Journal of Molluscan Studies 87.4, eyab041, 2021
- 2019 Fieldwork to sample microsnails for diet and microbiome studies along the Kinabatangan River, Sabah, Malaysian Borneo
K. P. Hendriks, ..., F. J. Richter Mendoza, M. Schilthuizen, R. S. Etienne
Malacologist 72, 33–38, 2019

INDUSTRY REPORTS

- 2024 IKEA: Innovative know-how for e-commerce allocation
Y. Chen, M. Pellemans, F. Richter, Y. Roshandel, V. Schmeits, P. Waldert, L. Wehbi, T. C. van der Zanden
SWI 2024 Scientific Proceedings
- 2024 Stereo 3D cloud motion from tandem satellites
Y. Zhou, S. Abrahams, F. Mendoza, M. Donà, R. Verbiest
Mathematics in Industry Reports

2023 – **Associate Editor** *Statistica Neerlandica*, Wiley

SELECTED TALKS

- 2026 **CCS — Conference on Complex Systems** Binghamton, NY, USA
Contributed oral presentation: How academic achievement and socioemotional learning are structurally linked — causal-discovery evidence from Peru's national assessment (with E. Aburto Camacllanqui).
- 2026 **EMS — European Meeting of Statisticians** Lugano, Switzerland
Speaker: The phylogenetic dark matter — Biostatistics session.
- 2026 **LACIAM — Latin American Congress on Industrial and Applied Mathematics** Univ. Técnica Federico Santa María, Valparaíso, Chile
Session organiser & speaker: Robust and Fair Prediction under Distribution Shift — thematic session *Algorithmic Fairness and Causal Modeling*.
- 2025 **Italian Statistical Society (SIS)** Genova, Italy
Invited talk: Neural Causal Regularization — Extending Causal Invariance to Deep Models.
- 2023 **Complex Networks (CompleNet)** Aveiro, Portugal
 Detecting complex diversity-dependent diversification with a novel phylogenetic inference framework.
- 2023 **World Conference on Natural Resource Modeling** Amsterdam, The Netherlands
 Phylodiversity-dependent diversification: a general framework for the role of phylogenetic diversity in diversification dynamics.
- 2022 **CCS — Conference on Complex Systems** Palma de Mallorca, Spain
 Inferring the drivers of species diversification processes.
- 2018 **CCS — Conference on Complex Systems** Thessaloniki, Greece
 Generalizing species diversification models.

OUTREACH

- 2025 – **TecDays** Secondary schools, Ticino
 Organised and delivered hands-on data-science workshops for high-school students across several schools.
- 2025 – **Lavori di Maturità** USI, Lugano
 Mentored high-school students through year-long thesis projects with the Faculty of Informatics.
- 2025 – **Giornate Autogestite** Ticino, Switzerland
 Ran workshops on data science and academic life at several high schools.